

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester IV Examination, 2013-14 (Old Course)

Zoology**Paper: ZOB-401 — Genetics, Evolution and Animal Behaviour**

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A, two questions from Section-B, and two questions from Section-C. Question No. 1 in all sections is compulsory. Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Genetics) (Marks: 36)**Question 1.** Answer the following parts:

- () Select the best option in the following: [2 × $\frac{1}{2}$ = 1]
- (i) If a homozygous tall plant is crossed with a heterozygous tall plant, which of the following is the most likely outcome for the progeny?
- (1) 100 tall : 0 dwarf
- (2) 25 tall : 50 dwarf : 25 tall
- (3) 75 tall : 25 dwarf
- (4) 67 tall : 33 dwarf
- (ii) In the ABO blood group system which of the following progeny is not possible?
- (1) An A child of A and B parents
- (2) An AB child of A and O parents
- (3) An O child of A and O parents
- (4) An O child of A parents
- () Write whether the following statements are True or False giving reasons in not more than 2–3 sentences: [2 × 3 = 6]
- (i) *Drosophila* having XY and XXY genotypes are males.
- (ii) A test cross is performed to find out the genotype of an individual.
- (iii) Kappa particles are involved in infective inheritance in *Paramecium*.
- () For the terms given in Column-A select the best match from those in Column-B: [4 × $\frac{1}{2}$ = 2]

Column-A

- (i) Cystic fibrosis
- (ii) Down syndrome
- (iii) Cry of cat syndrome
- (iv) CML

Column-B

- (1) 45, X
- (2) t(9; 22)
- (3) del 5p
- (4) CFTR

- () Differentiate between: [2 × 1½ = 3]
- (i) Dominance and epistasis
 - (ii) Penetrance and expressivity.

Question 2. Answer the following parts:

- () Define law of segregation and explain its chromosomal basis. [6]
- () How is G-banding performed and how does this banding help in karyotyping? How are the bands numbered on chromosomes? [6]

Question 3. Answer the following parts:

- () If an individual is heterozygous for a paracentric inversion, what will be the consequences following meiosis? [6]
- () While carrying out a dihybrid cross the scientist observed a F2 phenotypic ratio of 9:3:4. Explain the phenomenon. [6]

Question 4. Write notes on the following: [3 × 4 = 12]

- () Sex chromosome systems
- () Genetic counseling
- () Polygenic inheritance.

SECTION – B
(Evolution) (Marks: 17)

Question 1. Answer the following parts:

() Select the best option in the following: [2 × $\frac{1}{2}$ = 1]

(i) During which geological period gigantic dinosaurs, *Archaeopteryx* and early mammals coexisted?

- (1) Jurassic
- (2) Triassic
- (3) Cretaceous
- (4) Tertiary

(ii) The earliest ground dwelling hominid was

- (1) *Australopithecus*
- (2) *Dryopithecus*
- (3) *Ramapithecus*
- (4) *Homo erectus*

() Write whether the following statements are True or False giving reasons in not more than 2–3 sentences: [2 × 2 = 4]

- (i) If allelic frequency of A = 60% and that of a = 40% in a Mendelian population, then the genotype frequencies will be : AA = 48%, Aa = 16% and aa = 36%.
- (ii) Higher variation in the sequence of amino acids between different groups of organisms indicates closer phylogenetic relationship.

Question 2. Answer the following parts:

() Explain Natural selection in action, giving a suitable example. [6]

() Describe the major trends in human evolution. [6]

Question 3. Write notes on the following: [3 × 4 = 12]

- () Synthetic theory
- () Significance of homology and analogy in the evolutionary studies
- () Criticism on Lamarckism.

SECTION – C
(Animal Behaviour) (Marks: 17)

Question 1. Define the following:

[5 × 1 = 5]

- () Innate behaviour
- () Entrainment
- () Care-soliciting behaviour
- () Pheromones
- () Society.

Question 2. Explain the following:

[2 × 6 = 12]

- () Types of learning
- () Eusocial organization.

Question 3. Write notes on the following:

[4 × 3 = 12]

- () Circadian rhythms
- () Hormones and behaviour
- () Advantages of group living
- () Channels of communication.

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